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Pseudocode

Given Input
City_distances[] array
Gas_station[] array
Mpg int

Need # of cities, calculate using loop

```
num_cities
For x in city:
        Num_cities += 1
```

For main section need two loops to calculate algorithm

Inside loop starts at a city, then travels as far as possible
This is done via adding value in gas station array * mpg int and adding it to fuel
Then subtract # of miles to next city by fuel - city_distance to next city
Should only loop a number of times until it fails to travel to another city

Outside loop needs to change the starting city and reset the inside loops variables
Should only loop a number of times # to num cities max

Something along these lines

Loop for # of elements in city_distances or gas_station
{

```
        Starting_Pos = 0
        Current_Fuel =0
        Cities_travelled = 0
        Position = starting
                {
                Fuel = Fuel + (Gas_Station[Position]*mpg)
                Fuel = Fuel - (City_Distances[Position])
                Position = Position + 1
                //Used to loop when you reach the end
                If Position > # of elements in City_distances
                        Position = 0
                //Used before the loop check to preemptively increase the
                distance traveled
                If Fuel >= 0,
                        Cites_travelled +1
                Stop_loop = Stop_loop + 1
                }
        Starting_pos + 1
}
```

Big Oh Notation Using Step Count:

To intialize the project takes 5 steps

O(5)

First loop happens upwards of n times

O(n+5)

Outside loop has 5 step and occurs the same number of n times. Because its a nested loop, we will eventually multiply the two loops together

O(n+5+(5n*x))

Inside loop has upwards of 7 steps and can occur upwards of n times as well so we can fill the x with 7n

O(n+5+(5n(7n))

Simplifying the expression gives us
$O(n+5+(35n^2))=(35n^2+n+5)$
Simplifying it further to get the time complexity gives us a time complexity of
$O(n^2)$